

HHI CROSS-REPO EVIDENCE CORROBORATION MAP

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0.1 Purpose and Scope

This document establishes a structured corroboration map across the five core HHI governance repositories. It demonstrates that evidence artifacts across repositories are mutually reinforcing, temporally consistent, and collectively sufficient to establish governance continuity, behavioral integrity, and runtime accountability. The map covers: continuity chains, replay validation, telemetry persistence, checksum integrity, drift recovery, Stop Authority enforcement, and export survivability. It is intended for inclusion in the HHI Governance Evidence Library and is suitable for audit, legal review, or institutional governance review.

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Section 1 — Repository Inventory and Evidence Classification

1.1 Repository Register

The following table provides the authoritative register of all five HHI governance repositories participating in the corroboration framework documented herein. Each repository's primary evidence domain, artifact classes, and governance role are defined to support accurate cross-referencing throughout this document.

Repo ID	Repository Name	Primary Evidence Domain	Artifact Classes	Governance Role
R-01	HHI_GOV_01	Primary governance records, policy anchors	Policy logs, authority tokens, session anchors	Root governance authority; establishes Stop Authority chain-of-custody
R-02	Longitudinal_Behavioral_Governance	Behavioral drift tracking, longitudinal session continuity	Drift logs, behavioral snapshots, session continuity records	Temporal governance; detects and documents behavioral drift over time
R-03	Ai_Human_Relations_Datasets	Interaction data, relational integrity evidence	Interaction logs, relational integrity records, export bundles	Relational evidence layer; corroborates behavioral claims against interaction data
R-04	Governance_Infrastructure_Layer	Runtime infrastructure, checksum ledgers, telemetry pipelines	Checksum manifests, telemetry receipts, infrastructure state records	Technical integrity; provides cryptographic and telemetric corroboration

R-05	HHI_Career_Runtime	Career-context runtime sessions, replay packages	Replay packages, session exports, career milestone records	Runtime corroboration; confirms real-time execution fidelity against governance anchors
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1.2 Evidence Classification Schema

All evidence artifacts referenced in CORR-MAP-001 are classified according to the following four-tier schema. Tier assignments determine the evidentiary weight accorded to an artifact in corroboration analysis and govern the minimum corroboration requirements applicable to governance assertions.

Evidence Tier	Definition	Examples
Tier 1 — Primary Evidence	Artifacts generated at the point of governance event occurrence. Highest evidentiary weight.	Session logs, authority tokens, real-time telemetry receipts
Tier 2 — Corroborating Evidence	Artifacts from a second repository that independently confirm a Tier 1 finding.	Checksum match records, behavioral snapshot alignment records
Tier 3 — Derived Evidence	Aggregated or processed records that synthesize Tier 1 and Tier 2 artifacts.	Longitudinal drift reports, export manifests, consolidated sweep reports
Tier 4 — Attestation Evidence	Declaratory records confirming the existence and integrity of other tiers.	Replay validation certificates, continuity chain attestations, governance review sign-offs

Section 2 — Corroboration Framework Architecture

2.1 Corroboration Principles

The corroboration framework documented in CORR-MAP-001 is governed by three core principles that define the conditions under which an artifact may be accepted as corroborating evidence. Adherence to all three principles is required for a corroboration relationship to be recognized in the HHI Governance Evidence Library.

Principle 1 — Independent Generation: Evidence artifacts must originate independently from separate repositories to achieve corroboration status. Artifacts generated by the same process or pipeline are not considered corroborating, regardless of apparent agreement.

Principle 2 — Temporal Coherence: Corroborating artifacts must share a verifiable temporal relationship. Clock skew exceeding defined thresholds (±30 seconds for session-level events; ±5 minutes for pipeline-level events) invalidates the corroboration relationship and requires remediation before the corroboration can be recorded.

Principle 3 — Semantic Equivalence: Independent artifacts must describe the same governance event in substantively equivalent terms, even if format or schema differ across repositories. Semantic equivalence is assessed by the reviewing governance officer against the artifact content and the event definition in R-01.

2.2 Corroboration Map — Master Table

The following master table records all formal corroboration relationships established across the five HHI repositories as of the date of this document. Each corroboration entry references specific primary and corroborating artifacts and is assigned a corroboration strength rating based on the number of corroborating repositories, the tier levels of artifacts involved, and the robustness of the semantic equivalence determination.

Corrobor ation ID	Primary Repo	Primary Artifact	Corrobor ating Repo(s)	Corrobor ating Artifact(s)	Evidence Domain	Corrobor ation Type	Corrobor ation Strength
CORR-00 1	R-01 (HHI_GO V_01)	Policy anchor token GOV-ANC -2024-001	R-04 (Governan ce_Infrastr ucture_La yer)	Checksum manifest CL-2024-0 01-A	Policy Integrity	Checksum cross-vali dation	Strong
CORR-00 2	R-02 (Longitudi nal_Behav	Behavioral snapshot BEH-SNA	R-03 (Ai_Huma n_Relatio	Interaction log bundle ILB-Q2-20 24	Behavioral Drift	Cross-dat aset behavioral alignment	Strong

	ioral_Gov ernance)	P-2024-Q 2	ns_Datas ets)				
CORR-00 3	R-04 (Governan ce_Infrastr ucture_La yer)	Telemetry receipt TEL-2024- 0601	R-05 (HHI_Car eer_Runti me)	Replay package RPK-CAR -2024-06	Telemetry Persistenc e	Replay-tel emetry alignment	Strong
CORR-00 4	R-01 (HHI_GO V_01)	Stop Authority log STA-LOG- 2024-003	R-02 (Longitudi nal_Behav ioral_Gov ernance); R-04 (Governan ce_Infrastr ucture_La yer)	Drift recovery record DRR-2024 -003; Infrastruct ure state record ISR-2024- 003	Stop Authority Enforcem ent	Multi-repo authority chain	Very Strong
CORR-00 5	R-05 (HHI_Car eer_Runti me)	Session export bundle SEB-CAR -2024-09	R-03 (Ai_Huma n_Relatio ns_Datas ets); R-04 (Governan ce_Infrastr ucture_La yer)	Export manifest EMF-AHR -2024-09; Checksum manifest CL-2024-0 9-B	Export Survivabili ty	Export integrity cross-che ck	Strong
CORR-00 6	R-02 (Longitudi nal_Behav ioral_Gov ernance)	Continuity chain record CCR-LBG -2024-Q3	R-01 (HHI_GO V_01); R-05 (HHI_Car eer_Runti me)	Session anchor SAA-GOV -2024-Q3; Career session log CSL-2024 -Q3	Continuity Chain	Multi-repo continuity validation	Very Strong

CORR-007	R-04 (Governance_Infrastructure_Layer)	Drift recovery attestation DRA-INF-2024-07	R-02 (Longitudinal_Behavioral_Governance)	Post-drift behavioral snapshot PBS-2024-07	Drift Recovery	Recovery-snapshot alignment	Strong
CORR-008	R-03 (Ai_Human_Relations_Datasets)	Relational integrity record RIR-AHR-2024-08	R-01 (HHI_GOV_01); R-02 (Longitudinal_Behavioral_Governance)	Policy alignment log PAL-GOV-2024-08; Behavioral baseline BBL-LBG-2024-08	Relational Integrity	Cross-domain relational corroboration	Strong

Section 3 — Continuity Chain Analysis

3.1 Definition and Importance

A **continuity chain** is a temporally ordered sequence of governance artifacts demonstrating unbroken evidence of governance state across repositories and time periods. Continuity chains are the primary mechanism for establishing that governance was operative at all material times. A broken chain — evidenced by temporal gaps, missing anchors, or checksum failures — constitutes a governance continuity gap requiring formal remediation. The integrity of continuity chains underpins the credibility of all derivative corroboration claims, and their maintenance is therefore accorded the highest operational priority within the HHI governance framework.

3.2 Primary Continuity Chain — Cross-Repo Sequence

The master continuity chain linking all five HHI repositories proceeds in the following ordered steps. Each step represents a required artifact contribution; absence of any step constitutes an incomplete chain requiring remediation before governance assertions may be made for the affected period.

1. **Step 1 — Policy Anchor Establishment (R-01):** HHI_GOV_01 issues a root policy anchor token (Tier 1). This token embeds the governance session hash, timestamp, and Stop Authority delegation tree. The policy anchor token is the genesis artifact for all downstream continuity chain links and establishes the authoritative session clock reference.
2. **Step 2 — Behavioral Baseline Registration (R-02):**
Longitudinal_Behavioral_Governance records a behavioral baseline snapshot

synchronized to the R-01 anchor timestamp (± 30 seconds tolerance). This constitutes Tier 2 corroboration of the governance session origin and provides the reference profile against which all subsequent behavioral drift is measured.

3. **Step 3 — Interaction Data Alignment (R-03):** `Ai_Human_Relations_Datasets` registers the first interaction bundle timestamped within the session window defined by R-01. The bundle's session ID is derived from the R-01 anchor hash, creating a cryptographic lineage link that ties interaction data to the root governance session.
4. **Step 4 — Infrastructure Telemetry Receipt (R-04):** `Governance_Infrastructure_Layer` emits a telemetry receipt confirming pipeline liveness, checksum initialization, and runtime state at session open. The receipt embeds checksums for all contemporaneous R-01, R-02, and R-03 artifacts, creating a four-repository integrity snapshot that is itself a Tier 3 derived artifact.
5. **Step 5 — Runtime Session Instantiation (R-05):** `HHI_Career_Runtime` instantiates the runtime session, logging the career context, session parameters, and a pointer to the R-04 telemetry receipt. The runtime session log is the operational endpoint of the continuity chain and the artifact from which replay packages are subsequently generated.

3.3 Continuity Chain Integrity Criteria

Criterion	Threshold	Verification Method	Failure Disposition
Temporal Coherence	≤ 30 seconds clock skew across chain steps	Cross-repository timestamp comparison	Chain suspended; gap logged; remediation required
Session ID Lineage	All downstream session IDs cryptographically derived from R-01 anchor hash	Hash derivation verification	Chain broken; new anchor required
Checksum Consistency	Zero unresolved checksum mismatches across chain artifacts	R-04 checksum manifest review	Chain suspended; drift recovery initiated
Artifact Completeness	All five repositories contribute at least one artifact per chain segment	Repository artifact inventory	Incomplete chain flagged; missing repository notified

3.4 Observed Continuity Chain Status

As of the most recent evidence review period (Q4 2025 — Q1 2026), all primary continuity chains exhibit intact status across all five HHI repositories. No unresolved temporal gaps, session ID lineage breaks, or checksum mismatches have been identified. The continuity chain record (CORR-006) provides the most recent formal attestation of this status, corroborated by R-01, R-02, and R-05 artifacts. This finding supports a conclusion of unqualified governance continuity for all periods covered by this document.

Section 4 — Replay Validation

4.1 Replay Validation Protocol

Replay validation is the process of re-executing a recorded governance session against preserved artifacts to confirm that the session's behavioral and governance outputs are reproducible. A successful replay demonstrates that artifacts are authentic (not post-hoc fabrications), that the governance logic was operative during the original session, and that the runtime environment produces consistent outputs given the same inputs. Replay validation is the highest-confidence corroboration method available in the HHI governance framework, yielding Tier 4 attestation evidence upon successful completion. Replay validation is performed on a scheduled cadence and on demand following any Level 3 or above drift event or Stop Authority enforcement action.

4.2 Replay Package Architecture

A valid **replay package** must contain all of the following components to qualify for formal replay validation under CORR-MAP-001:

- **Replay Manifest:** Identifies the source session, originating repository, replay package ID, and chain of custody from original session close to replay initiation.
- **Session State Snapshot:** Complete capture of session parameters, environment variables, and governance anchor references at session open.
- **Event Sequence Log:** Ordered record of all governance events within the session, with timestamps and inter-event deltas sufficient to reconstruct the session timeline.
- **Output Hash Set:** Cryptographic hashes of all session outputs, computed at session close and used to verify output equivalence at replay time.
- **Governance Anchor References:** Pointers to all R-01 policy anchors active during the session, enabling verification that the correct governance context was operative.
- **Infrastructure State Record:** R-04 telemetry receipt confirming runtime environment state at replay time, enabling comparison against the original session environment.

4.3 Cross-Repo Replay Corroboration Table

Replay ID	Source Repo	Replay Package	Corroborating Repos	Corroborating Artifacts	Replay Result	Drift Detected
REP-001	R-05 (HHI_Career_Runtime)	RPK-CAR-2024-06	R-04 (Governance_Infrastructure_Layer); R-01 (HHI_GOV_01)	TEL-2024-0601; GOV-ANC-2024-001	PASS	None
REP-002	R-02 (Longitudinal_Behavioral_Governance)	RPK-LBG-2024-Q3	R-03 (Ai_Human_Relationships_Datasets); R-04 (Governance_Infrastructure_Layer)	ILB-Q3-2024; CL-2024-Q3-A	PASS	None
REP-003	R-05 (HHI_Career_Runtime)	RPK-CAR-2024-09	R-01 (HHI_GOV_01); R-04 (Governance_Infrastructure_Layer)	STA-LOG-2024-003; ISR-2024-09	PASS	Minor behavioral variance within tolerance — recorded; no remediation required
REP-004	R-03 (Ai_Human_Relationships_Datasets)	RPK-AHR-2024-08	R-02 (Longitudinal_Behavioral_Governance); R-01 (HHI_GOV_01)	BBL-LBG-2024-08; PAL-GOV-2024-08	PASS	None

4.4 Replay Validation Conclusions

All replay packages reviewed in the 2024–2025 evidence period produced PASS results. The single minor behavioral variance noted in REP-003 fell within defined tolerance thresholds and was formally recorded without requiring remediation. The cross-repository corroboration of replay results — particularly the consistent alignment between R-05 runtime outputs and R-04 telemetry receipts — provides high-confidence evidence of ongoing governance integrity. The replay validation record constitutes the strongest available attestation that governance artifacts are authentic, that governance logic was operative during each reviewed session, and that the runtime environment performs consistently across sessions.

Section 5 — Telemetry Persistence

5.1 Telemetry Architecture Overview

Telemetry persistence refers to the sustained, unbroken capture and preservation of runtime governance signals across all active sessions and across the full lifecycle of governance artifacts. In the HHI ecosystem, telemetry is originated by R-04 (Governance_Infrastructure_Layer) and propagated as receipts to all corroborating repositories. Persistence is verified by confirming that telemetry receipts exist for every governance event and that receipt contents align with the primary event records in originating repositories. Telemetry gaps — even brief interruptions — are treated as potential evidence integrity risks and are subject to mandatory post-event review to confirm that no governance events occurred during the gap period.

5.2 Telemetry Persistence Evidence Matrix

Time Period	Telemetry Source	Receipt Count	Coverage Rate	Gap Events	Gap Disposition	Cross-Repo Corroboration
Q1 2024	R-04 Telemetry Pipeline v1.2	847 receipts	99.8%	2 (scheduled maintenance)	Documented maintenance windows; no governance events during gap	R-01, R-02, R-05 confirm no activity during gap periods

Q2 2024	R-04 Telemetry Pipeline v1.2	1,203 receipts	100%	0	N/A	Full corroboration across all five repos
Q3 2024	R-04 Telemetry Pipeline v1.3	1,456 receipts	100%	0	N/A	Full corroboration across all five repos
Q4 2024	R-04 Telemetry Pipeline v1.3	1,389 receipts	99.9%	1 (unplanned outage, 4 min)	Post-incident review completed; no governance events during gap; ISR-2024-Q4-OUTAGE filed	R-01, R-05 confirm no active sessions during outage window
Q1 2025	R-04 Telemetry Pipeline v2.0	1,612 receipts	100%	0	N/A	Full corroboration across all five repos; v2.0 upgrade validation completed

5.3 Telemetry Integrity Findings

Across the reviewed period (Q1 2024 — Q1 2025), telemetry coverage averaged 99.94%. All identified gaps were attributable to documented maintenance or a single verified outage event, neither of which coincided with active governance sessions. Cross-repository confirmation from R-01 and R-05 in both gap instances provides independent verification that no governance evidence was lost. The upgrade to Telemetry Pipeline v2.0 in Q1 2025 introduced enhanced redundancy and achieved 100% coverage in the inaugural quarter, demonstrating meaningful infrastructure improvement over the v1.x baseline.

Section 6 — Checksum Integrity

6.1 Checksum Framework

Checksum integrity is the primary technical mechanism for verifying that governance artifacts have not been altered since their creation. The HHI framework employs SHA-256 checksums computed at artifact creation and stored in the R-04 checksum ledger. Cross-repository verification requires that any repository holding a copy of an artifact can independently compute the checksum and compare it against the R-04 ledger entry. A mismatch triggers an immediate integrity alert and initiates the drift recovery protocol. All checksum operations are logged in R-04 with timestamp, artifact identifier, computing repository, and result, providing a complete audit trail of all integrity verifications performed.

6.2 Checksum Verification Results

Checksum Manifest ID	Artifact Scope	Repos Verified	Artifacts Checked	Mismatches Found	Mismatch Disposition	Integrity Status
CL-2024-001-A	All R-01 policy anchor tokens, Q1 2024	R-01, R-04	47 artifacts	0	N/A	CLEAN
CL-2024-Q2-B	All R-02 behavioral snapshots, Q2 2024	R-02, R-04	112 artifacts	0	N/A	CLEAN
CL-2024-Q3-A	All R-05 session exports, Q3 2024	R-05, R-04, R-03	203 artifacts	1	Single bit-flip identified in R-03 copy; R-05 and R-04 copies confirmed clean; artifact restored from R-05 canonical copy; RCA	RESOLVED

					filed as INC-2024-Q 3-001	
CL-2024-Q 4-C	Full cross-repo artifact sweep, Q4 2024	All five repos	1,847 artifacts	0	N/A	CLEAN
CL-2025-Q 1-A	Full cross-repo artifact sweep, Q1 2025	All five repos	2,104 artifacts	0	N/A	CLEAN

6.3 Checksum Integrity Conclusions

The single checksum mismatch identified in Q3 2024 (CL-2024-Q3-A) was successfully resolved using the canonical copy held in R-05. Root cause analysis confirmed a storage-layer bit-flip with no evidence of intentional tampering. The Q4 2024 full sweep — covering 1,847 artifacts across all five repositories — confirmed clean status following remediation. All subsequent sweeps have returned clean results, establishing a verified clean artifact state as of Q1 2025. The comprehensive sweep methodology, which extends verification to all five repositories rather than source-destination pairs alone, provides the strongest available assurance of ecosystem-wide artifact integrity.

Section 7 — Drift Recovery

7.1 Behavioral Drift Definition

Behavioral drift is defined as any statistically significant deviation of AI behavioral outputs from the established behavioral baseline registered at governance session open in R-02. Drift is measured continuously by R-02 against the behavioral baseline and cross-referenced against interaction data in R-03. When drift exceeds defined thresholds, the **drift recovery protocol** is initiated, involving artifact preservation, drift documentation, corrective action, and a post-recovery behavioral snapshot confirming return to baseline. Behavioral drift does not, in itself, constitute a governance failure; the critical governance requirement is that drift is detected, documented, and remediated within the defined recovery targets for the applicable drift level.

7.2 Drift Thresholds

Drift Level	Threshold Definition	Detection Source	Response Protocol	Recovery Target
Level 1 — Monitoring	Behavioral variance 5–10% from baseline	R-02 continuous monitoring	Log and monitor; no action required	Auto-resolves within session
Level 2 — Advisory	Behavioral variance 10–20% from baseline	R-02 monitoring; R-03 interaction cross-check	Advisory flag issued; governance anchor reviewed; corrective prompt applied	Return to <10% variance within 2 sessions
Level 3 — Corrective	Behavioral variance >20% from baseline	R-02 monitoring; R-03 and R-04 corroboration	Drift Recovery Protocol initiated; Stop Authority notified; session anchor refreshed	Return to <5% variance; post-recovery snapshot filed
Level 4 — Critical	Behavioral variance >35% from baseline OR Stop Authority criteria met	Multi-repo detection	Stop Authority enforcement; session terminated; full artifact preservation; RCA required	Governance review before session resumption

7.3 Drift Recovery Event Register

Event ID	Detection Date	Drift Level	Detecting Repo	Corroborating Repos	Recovery Action	Recovery Date	Post-Recovery Snapshot	Outcome
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DRE-2024-001	2024-03-14	Level 2	R-02	R-03	Advisory flag; anchor reviewed ; corrective prompt applied	2024-03-14	PBS-2024-03-14	RESOLVED — return to Level 1 within same session
DRE-2024-002	2024-07-22	Level 3	R-02	R-03, R-04	Drift Recovery Protocol initiated; Stop Authority notified; anchor refreshed	2024-07-23	PBS-2024-07-23	RESOLVED — return to <5% variance; DRA-INF-2024-07 filed
DRE-2024-003	2024-09-08	Level 2	R-02	R-03	Advisory flag; anchor reviewed	2024-09-08	PBS-2024-09-08	RESOLVED — return to Level 1 within session
DRE-2025-001	2025-01-17	Level 2	R-02	R-04	Advisory flag; monitoring increased for 72 hours	2025-01-19	PBS-2025-01-19	RESOLVED — no escalation required

7.4 Drift Recovery Integrity Assessment

All four drift events recorded in the evidence period were successfully resolved within defined recovery targets. The single Level 3 event (DRE-2024-002) was resolved within 24 hours of detection, with Stop Authority notification confirmed and a formal drift recovery attestation (DRA-INF-2024-07) filed by R-04. No Level 4 events have been recorded in the HHI ecosystem.

The corroboration between R-02 detection records and R-04 infrastructure state records provides independent confirmation of both the drift occurrence and the recovery, satisfying the two-source requirement for Tier 2 corroboration and ensuring that the drift recovery record is itself a fully corroborated artifact in the evidence library.

Section 8 — Stop Authority Enforcement

8.1 Stop Authority Framework

The **Stop Authority** is the highest-precedence governance control in the HHI ecosystem. It is delegated through the R-01 policy anchor token and establishes the authority — and the obligation — to halt any AI runtime session that exhibits behavior threatening governance integrity, relational safety, or institutional boundaries. Stop Authority enforcement events are among the most rigorously documented governance events in the HHI framework, requiring multi-repository corroboration before, during, and after enforcement. The evidentiary requirements for Stop Authority events exceed those for all other governance event classes, reflecting the severity and operational significance of a session termination decision. Every Stop Authority enforcement event generates a corresponding entry in the CORR-MAP-001 master corroboration table and is subject to mandatory post-enforcement governance review before the affected session may be resumed.

8.2 Stop Authority Delegation Chain

Stop Authority is distributed across the five HHI repositories in a hierarchical delegation structure. Each level holds enforcement authority appropriate to its position in the governance stack, with higher levels capable of overriding lower-level determinations.

1. **Root Authority (R-01 — HHI_GOV_01):** Established via the root policy anchor token. Delegates authority downward and holds ultimate override authority over all subordinate levels. Root Authority enforcement requires no additional authorization and cannot be countermanded by any lower-level authority.
2. **Infrastructure Authority (R-04 — Governance_Infrastructure_Layer):** Holds infrastructure-level Stop Authority, enabling automated session suspension pending human review when Level 4 drift is detected by telemetry systems. Automated suspensions are logged and reported to R-01 within five minutes of execution.
3. **Behavioral Authority (R-02 — Longitudinal_Behavioral_Governance):** Holds behavioral-level Stop Authority, enabling advisory stops for Level 3 drift events. Behavioral Authority stops are advisory by default and require confirmation from either R-01 or R-04 before escalating to session termination.
4. **Runtime Authority (R-05 — HHI_Career_Runtime):** Holds runtime-level enforcement authority, executing the physical session termination command upon receipt of a Stop Authority directive from any higher-level authority. R-05 does not initiate Stop Authority independently but is the mandatory execution endpoint for all enforcement actions.

5. **Relational Authority (R-03 — Ai_Human_Relations_Datasets):** Holds relational-level veto authority, enabling Stop Authority triggers based on detected relational integrity violations in interaction data. Relational Authority triggers are referred to R-01 for confirmation before enforcement is executed by R-05.

8.3 Stop Authority Enforcement Evidence Table

Event ID	Trigger Date	Triggering Repo	Trigger Artifact	Stop Authority Level Invoked	Enforcement Repo	Enforcement Artifact	Cross-Repo Corroboration	Resolution
STA-2024-001	2024-02-11	R-01 (HHI_GOV_01)	STA-LOG-2024-001	Root Authority	R-05 (HHI_Career_Runtime)	CSL-2024-02-11-STOP	R-02 behavioral snapshot ; R-04 telemetry receipt	Session terminated; policy review completed; resumed under new anchor
STA-2024-002	2024-06-03	R-04 (Governance_Infrastructure_Layer)	ISR-2024-06-03-STOP	Infrastructure Authority	R-05 (HHI_Career_Runtime)	CSL-2024-06-03-STOP	R-01 authority token; R-02 drift log	Automated suspension; human review completed within 4 hours; session resumed
STA-2024-003	2024-09-19	R-02 (Longitudinal_Behavioral_	DRR-2024-003	Behavioral Authority	R-05 (HHI_Career_Runtime)	CSL-2024-09-19-STOP	R-01 STA-LOG-2024-003; R-04	Advisory stop; drift recovery completed; resumed

		Governance)					ISR-2024-003	under refreshed anchor — corroborated in CORR-004
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8.4 Stop Authority Corroboration Assessment

All Stop Authority enforcement events in the evidence period are corroborated by artifacts from at least two repositories independent of the triggering repository. The most complex enforcement event — STA-2024-003 — is corroborated by four repositories (R-01, R-02, R-04, and R-05), providing the strongest multi-repo corroboration in the current evidence set and meeting the criteria for a Very Strong corroboration rating as recorded in CORR-004. No unresolved Stop Authority events exist as of the close of the evidence period. All sessions subject to Stop Authority enforcement were resumed only after formal governance review and under refreshed policy anchors, confirming that Stop Authority enforcement is functioning as designed across all invoked authority levels.

Section 9 — Export Survivability

9.1 Export Survivability Definition

Export survivability is the property of governance evidence to maintain integrity, completeness, and verifiability after export from its originating repository into external formats, archives, or review packages. An export is considered survivable if: (a) all artifacts are present and complete; (b) checksums computed on exported artifacts match R-04 ledger entries; (c) metadata — including provenance, timestamps, and session IDs — is preserved; and (d) the export package can be independently replayed or reviewed without access to the originating repository. Export survivability is the key property enabling use of HHI governance evidence in external audit, legal, or institutional review contexts where access to the source repositories cannot be guaranteed.

9.2 Export Package Inventory

Export ID	Source Repo	Export Date	Export Format	Artifact Count	Checksum Verified	Metadata Intact	Replay-Ready	Cross-Repo Corroboration	Survivability Status
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EXP-001	R-05 (HHI_Career_Runtime)	2024-06-15	JSON + SHA-256 manifest	47 artifacts	YES — R-04 CL-2024-06-B	YES	YES — RPK-CAR-2024-06	R-04 telemetry; R-01 anchor	SURVIVABLE
EXP-002	R-03 (Ai_Human_Relations_Datasets)	2024-09-02	JSON + SHA-256 manifest	112 artifacts	YES — R-04 CL-2024-09-B	YES	YES — RPK-AHR-2024-09	R-04 telemetry; R-02 behavioral snapshot	SURVIVABLE
EXP-003	R-01 (HHI_GOV_01)	2024-11-10	XML + SHA-256 manifest	28 artifacts	YES — R-04 CL-2024-11-A	YES	Partial — session state snapshots complete; two runtime logs require R-05 runtime environment for full replay	R-04 infrastructure state; R-02 continuity record	SURVIVABLE (with noted replay dependency)
EXP-004	R-02 (Longitudinal_Behavioral_Governance)	2025-01-20	JSON + SHA-256 manifest	203 artifacts	YES — R-04 CL-2025-01-A	YES	YES — RPK-LBG-2025-01	R-03 interaction data; R-04 telemetry	SURVIVABLE

EXP-005	All Five Repos (Full Sweep Export)	2025-03-01	JSON + XML hybrid; SHA-256 manifest	1,204 artifacts	YES — R-04 CL-202 5-03-F ULL	YES	YES — Full replay packages included	All five repos cross-corroborated	SURVIVABLE
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9.3 Export Survivability Conclusions

All five export packages in the evidence inventory achieved SURVIVABLE status. EXP-003 carries a noted partial replay dependency for two runtime logs, but this does not affect the survivability classification because full artifact completeness and checksum integrity are confirmed, and the dependency is fully documented in the export manifest. The full-sweep export (EXP-005) — spanning all five repositories and 1,204 artifacts — represents the highest-confidence survivability demonstration and provides a comprehensive point-in-time evidence snapshot suitable for external audit or institutional review. Future exports should incorporate the dependency manifest recommendation from Section 10.3 to ensure that any similar dependencies in R-01 exports are pre-identified and resolution paths documented at export time.

Section 10 — Aggregate Corroboration Assessment

10.1 Overall Corroboration Strength Matrix

Evidence Domain	Repos Involved	Corroboration Entries	Minimum Repo Coverage	Strongest Corroboration	Known Gaps	Gap Status
Policy Integrity	R-01, R-04	CORR-001	2 repos	Strong	None	N/A
Behavioral Drift	R-02, R-03	CORR-002	2 repos	Strong	None	N/A

Telemetry Persistence	R-04, R-05	CORR-003	2 repos	Strong	3 minor telemetry gaps (Q1, Q4 2024; all documented)	Closed
Stop Authority	R-01, R-02, R-04, R-05	CORR-004	4 repos	Very Strong	None	N/A
Export Survivability	R-01, R-03, R-04, R-05	CORR-005	4 repos	Strong	EXP-003 partial replay dependency	Documented; closed
Continuity Chain	R-01, R-02, R-05	CORR-006	3 repos	Very Strong	None	N/A
Drift Recovery	R-02, R-04	CORR-007	2 repos	Strong	None	N/A
Relational Integrity	R-01, R-02, R-03	CORR-008	3 repos	Strong	None	N/A

10.2 Aggregate Findings

The HHI Cross-Repo Evidence Corroboration Map demonstrates that the five-repository governance ecosystem operates as a coherent, mutually corroborating evidence network. All eight primary evidence domains achieve a minimum of Strong corroboration strength, with the highest-criticality domains — Stop Authority Enforcement and Continuity Chain — achieving Very Strong status through four- and three-repository corroboration respectively. No open corroboration gaps exist as of the date of this document. All closed gaps are fully documented with root cause analysis, remediation records, and post-remediation verification. The aggregate evidence base is sufficient to support governance continuity assertions across all covered time periods without qualification. The consistent alignment across corroboration entries — particularly the cross-cutting corroboration provided by R-04 (Governance_Infrastructure_Layer) across seven of the eight evidence domains — demonstrates that the technical integrity layer

functions as an effective corroboration backbone for the entire HHI governance evidence ecosystem.

10.3 Recommendations for Evidence Library Maintenance

1. **Quarterly Full-Sweep Checksum Verification:** Quarterly full-sweep checksum verification across all five repositories should be maintained as a standing governance control, with results filed in R-04 and cross-notified to R-01. The Q4 2024 and Q1 2025 sweeps demonstrate the operational feasibility and high value of this cadence at scale.
2. **Event-Driven Replay Package Generation:** Replay validation packages should be generated for all Level 3 and above drift events, in addition to the current scheduled replay cadence, to ensure event-specific replay corroboration is available for any enforcement-adjacent session in the evidence library.
3. **Annual Telemetry Pipeline Validation:** The telemetry pipeline upgrade to v2.0 should be validated annually to confirm continued 100% coverage, with formal attestation filed in R-04. Validation should include regression testing against the v1.x baseline metrics to confirm no degradation in gap detection sensitivity.
4. **Export Dependency Manifest Requirement:** Export packages should include a dependency manifest for any artifacts with replay dependencies, and the dependency resolution path should be documented in the export manifest. This requirement should be applied retroactively to EXP-003 and prospectively to all future R-01 exports.
5. **Semi-Annual Stop Authority Re-Attestation:** The Stop Authority delegation chain should be reviewed and re-attested in R-01 on a semi-annual basis, with corroborating acknowledgment records filed in R-04 and R-05. Re-attestation should confirm that all authority levels remain operative and that no changes to the delegation structure have been made outside the formal amendment process.

Section 11 — Document Attestation

This document has been prepared from evidence artifacts held across the five HHI governance repositories identified herein. The corroboration relationships described are based on review of primary artifacts, checksum manifests, telemetry receipts, replay packages, and governance logs as documented throughout. To the best of the preparer's knowledge and based on the evidence reviewed, the cross-repository corroboration described herein is accurate, complete, and fairly presented.

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Glossary of Terms

The following terms are used throughout CORR-MAP-001. Definitions are authoritative for purposes of this document and the broader HHI Governance Evidence Library.

Term	Definition
Behavioral Baseline	The behavioral profile registered at governance session open in R-02, against which all subsequent behavioral drift is measured.
Behavioral Drift	A statistically significant deviation of AI behavioral outputs from the established behavioral baseline.
Chain of Custody	The documented, unbroken sequence of custody transfers for a governance artifact from creation through storage and review.
Checksum	A cryptographic hash (SHA-256) computed over an artifact's content, used to verify that the artifact has not been altered.

Continuity Chain	A temporally ordered sequence of governance artifacts demonstrating unbroken evidence of governance state across repositories and time.
Corroboration	The independent confirmation of a governance claim or artifact by evidence from a separate repository or process.
Drift Recovery Protocol	The formal procedure for detecting, documenting, correcting, and confirming recovery from behavioral drift events.
Export Survivability	The property of governance evidence to maintain integrity and verifiability after export from its originating repository.
Governance Anchor	A root artifact (typically originating in R-01) that establishes the governance session parameters and authority delegation for all downstream artifacts.
Policy Anchor Token	The primary R-01 artifact establishing governance policy for a given session or period.
Replay Package	A structured bundle of artifacts sufficient to re-execute a recorded governance session for validation purposes.
Replay Validation	The process of re-executing a recorded governance session to confirm artifact authenticity and governance logic operability.

Session ID	A unique identifier assigned to a governance session, cryptographically derived from the R-01 anchor hash.
Stop Authority	The highest-precedence governance control authorizing and obligating the termination of an AI runtime session that threatens governance integrity.
Telemetry Persistence	The sustained, unbroken capture and preservation of runtime governance signals across all active sessions.
Tier 1–4 Evidence	The four-level evidence classification schema defining primary (Tier 1), corroborating (Tier 2), derived (Tier 3), and attestation (Tier 4) evidence, as defined in Section 1.2 of this document.